Project Proposal

CSE 4104: Software Development III | AI-Based Software Project

Team Name	CSE4104-7B-T01
Section	7B
Project Title	CareBridge Khulna
GitHub Repository	https://github.com/114nahid/cse4104-7b-t01-carebridge-khulna

1. Project Title

CareBridge Khulna: An AI-Powered Healthcare Assistance Platform for Elderly People

CareBridge Khulna is a web application that helps families take care of the medical needs of their elderly parents in Khulna city, even when they live far away. Families can find the right doctor, book a trusted university student to take the patient to the hospital and back home, and get updates at every step of the visit.

2. Problem Statement

What is the problem?

Many elderly people in Khulna cannot go to the doctor alone. Some have walking problems, some cannot handle the hospital process (serial, tests, reports), and some do not feel safe going alone. At the same time, many of their children live in Dhaka, other cities or abroad for work or study. Because of this, elderly patients often miss their checkups or go alone with a lot of difficulty.

Why is this problem important?

- Elderly people need regular checkups for problems like diabetes, blood pressure and heart disease.
- If treatment is late, a small problem can become serious.
- Family members stay worried because they cannot be there and do not know what happened at the hospital.

Why existing solutions are not enough

- Most existing apps and websites only help to book a doctor's serial or appointment.
- Nobody goes with the patient to the hospital.
- Family members do not get any update during the visit.

- Families often do not know which specialist doctor they should choose for a health problem.
- Prescriptions are hard to read and understand, especially for family members who live far away.

Why AI is useful here

- AI can understand health problems written or spoken in Bangla and suggest the right type of specialist doctor.
- AI can read a prescription photo and explain the medicines and timing in simple Bangla.
- AI can turn the student assistant's short notes into a clear visit report for the family.
- AI can notice emergency signs (for example, "fainted" or "chest pain") and quickly show emergency help.

3. Objectives

- To help families find the right specialist doctor in Khulna from anywhere.
- To provide verified university students who can take elderly patients to the hospital and bring them back home safely.
- To keep family members updated at every step of the hospital visit.
- To use AI to suggest specialists, explain prescriptions and create visit reports.
- To build a simple and easy-to-use system that works well on both mobile and computer.

4. Proposed Features

For Family Members

- Sign up and log in
- Add elderly parent's profile (age, address, health conditions)
- Search doctors by specialty and area
- Ask AI by describing the health problem
- Search by voice in Bangla
- Pin the patient's home on a map and see the doctor's chamber location
- Book hospital assistance for a date and doctor
- See visit status step by step
- Read prescription summary and visit report
- Rate and review the student assistant

For Student Assistants

- Register with university ID card and NID
- Start working after admin approval
- Accept assigned visits
- Update visit status step by step
- Upload prescription photo and write short notes after the visit

For Admin

- Verify student assistants
- Add and manage doctor information
- Assign students to bookings
- Monitor all bookings and reviews

AI Features

Feature	What it does
AI Specialist Finder	The user writes or speaks the problem (for example, "my father feels chest pressure when walking"). AI suggests the right type of specialist (Cardiologist) and shows matching doctors in Khulna.
AI Prescription Summary	Reads the prescription photo and shows medicine names, when to take them and the next visit date in simple Bangla.
AI Visit Report	Turns the student's short notes into a clear report for the family, in Bangla or English.
Emergency Alert	If the message shows emergency signs, the system shows the 999 number and nearby emergency departments right away.

Other Features

- Notifications inside the website for bookings and status updates
- Separate dashboard for each type of user
- Mobile-friendly design

5. Technology Stack

Part	Technology	Use in our project
Frontend	React.js + Tailwind CSS	User interface and dashboards
Backend	Node.js + Express.js	REST APIs, booking logic, role management
Database	PostgreSQL	Users, doctors, bookings, visit status, reviews
Authentication	JWT + bcrypt	Secure login and password hashing
AI Integration	Gemini API	Specialist finder, prescription summary, visit report
Voice Assistant	Web Speech API	Bangla voice search
Image Storage	Cloudinary	Prescription photos and student ID images
Map	Leaflet + OpenStreetMap, OpenRouteService	Patient home location, doctor chamber location, distance to hospital
Deployment	Vercel, Render, Supabase	Frontend, backend and database hosting
Tools	GitHub, Figma, Postman, Draw.io	Version control, UI design, API testing, diagrams

6. Expected Outcome

What users will gain

- Families can arrange their parents' treatment from anywhere and stay worry-free.
- Elderly patients do not have to go to the hospital alone.
- Families get updates during the visit and a clear report after it.
- University students get a part-time earning opportunity while helping people.

How it improves the current way

Now	With CareBridge Khulna
A family member has to travel, or ask a neighbour or relative for help	Book a verified student online from anywhere
The family does not know which doctor to choose	AI suggests the right specialist
No update during the hospital visit	Step-by-step status updates
The prescription is hard to understand	AI explains it in simple Bangla
The family hears about the visit only by phone call	A clear written visit report after every visit

Benefits of AI integration

- Saves time in finding the right doctor.
- Makes medical information easy to understand for common people.
- Gives the family a clear report without extra work for the student.
- Helps to respond faster in an emergency.

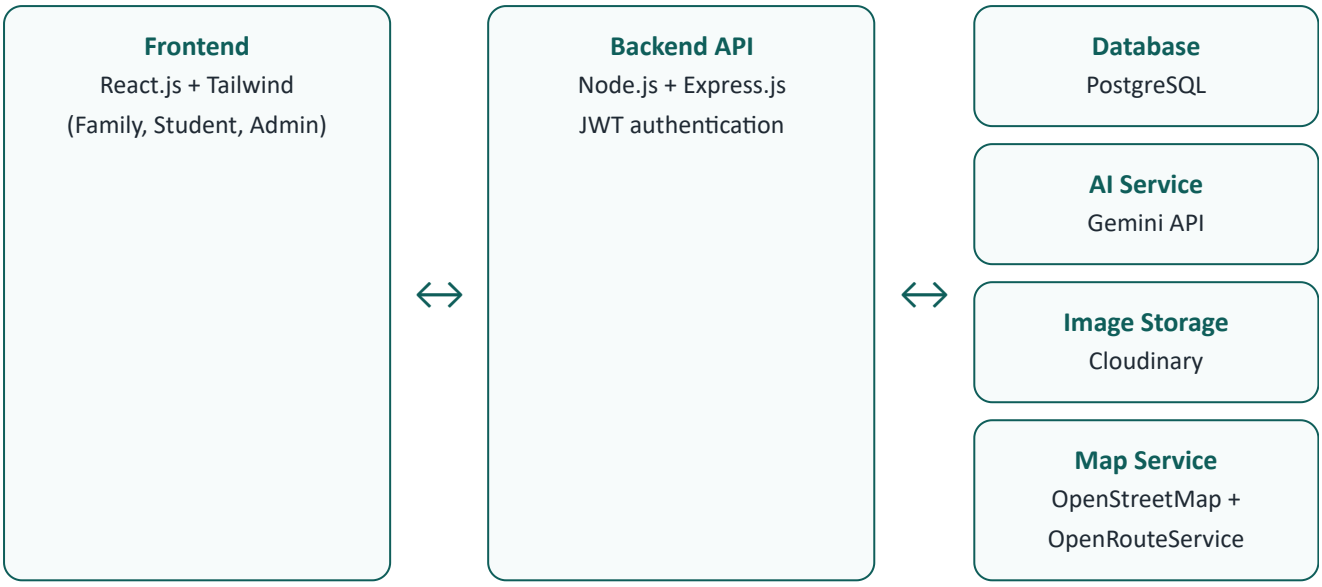
7. How the System Works

Hospital visit flow

Family books a visit → Admin assigns a verified student → Student picks up the patient → Reached hospital → With doctor → Prescription uploaded → AI summary and report → Patient back home

The family can see every step on their dashboard.

System architecture



8. Project Scope

We will build

- Login system with three user roles
- Doctor directory and AI specialist finder
- Hospital assistance booking with map location
- Student verification and assignment
- Step-by-step visit status
- AI prescription summary and visit report
- Ratings, reviews and notifications

We will not build in this semester

- Live GPS tracking
- Online payment
- Direct connection with hospital systems (the appointment will be confirmed by phone)
- Separate login for doctors
- Mobile app (the website will be mobile-friendly)

Safety note: AI will not diagnose any disease or suggest any medicine. It will only suggest the type of specialist and explain the doctor's prescription in simple words. This will be clearly shown in the app.

9. Initial Project Plan

Week	Work	Deliverable
1	Team formation and idea selection	Team info, project idea
2	Project proposal	Proposal, GitHub repository
3	Requirement analysis	SRS document
4	System design	Architecture, ER diagram, API list
5	UI/UX design	Figma design
6	Backend development	Login, database, APIs
7	Frontend development	Pages connected with APIs
8	AI integration	Specialist finder, prescription summary, visit report
9	Feature completion	Nearly complete system
10	Testing and bug fixing	Testing report
11	Deployment	Live website link
12	Documentation	Final report, user manual, API documentation
13	Presentation preparation	Presentation slides
14	Final presentation and viva	Live demo

10. Team Members and Responsibilities

Name	Student ID	Email	Responsibility
M Sajjad Hossain *(Team Leader)*	11230121108	sajjad.cse07@gmail.com	Database (PostgreSQL), AI integration (Gemini API), documentation and team coordination
Nahid Hasan Sabbir	11250122021	114nahidhasan@gmail.com	Backend (Node.js + Express.js), REST APIs and authentication (JWT)
Md Sohanujjaman Sheikh	11230321265	mdrafin2002@gmail.com	Frontend (React.js + Tailwind CSS) and UI/UX design